

CoRoT-1b

R-2662

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-1_130111 · created 2026-08-30T00:30:11+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2456303.6992 +/- 0.002 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.14 +/- 0.04
Transit depth (Rp/Rs)^2	1.96 +/- 1.13 [%]
Orbital Inclination (inc)	89.36 +/- 2.83
Airmass coefficient 1 (a1)	0.49 +/- 0.28
Airmass coefficient 2 (a2)	0.2 +/- 0.43
Scatter in the residuals of the lightcurve fit is	54.56 %
Best Comparison Star	#3 - [174, 36]
Optimal Aperture	2.05
Optimal Annulus	7.5
Transit Duration (day)	0.1118 +/- 0.0059

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.14 ± 0.04 vs 0.1388 (deviation 0.03σ)
Duration fit vs published	0.1118 d vs 0.1046 d (deviation 1.22σ)
Mid-time O–C	2.2 min
Depth SNR	3.5
Residual scatter	54.56 %

Light curve

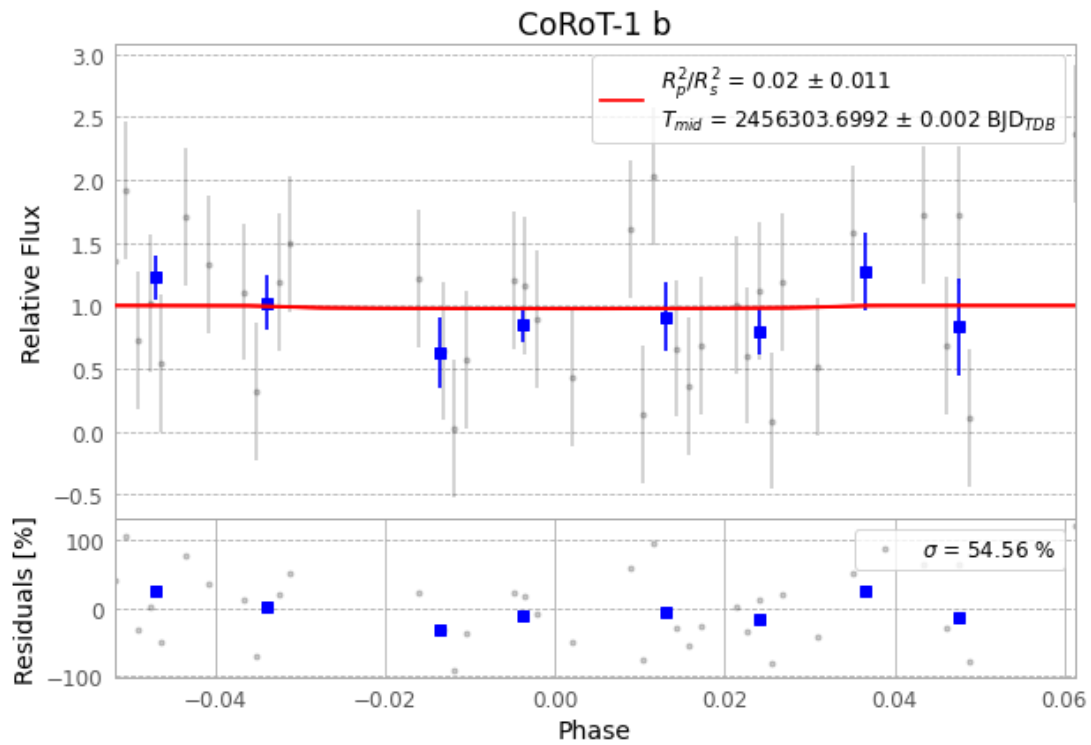


Figure 1 — FinalLightCurve_CoRoT-1 b_2013-01-10.png (SHA-256 c7c0cf6ddeb4758c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [582, 134], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 029aa93cff262ac2...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

86 FITS frames (56 MB), CoRoT-1130111023912.FITS → CoRoT-1130111065412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-1-130111.log (43c9c9f7a9ec...), FinalLightCurve_CoRoT-1 b_2013-01-10.png (c7c0cf6ddeb4...), FinalLightCurve_CoRoT-1 b_2013-01-10.csv (adfa551f20ba...)

Compute resources

Wall time 123.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-30 00:30Z.